

ApniLeap — Demo Data Cheat Sheet

Pre-Validated Sample **KPIs** and **Challenges** for Live Portal & Jira Demonstration

Demo Quick Tip: Copy and paste the values below directly into the ApniLeap web interface. Adding a **Challenge** automatically creates a corresponding issue on Jira Cloud Kanban board. Adding a **KPI** and recording measurements updates the live completion metrics and project tracking.

Project: AL-KLE-026 — Smart Campus Lab & Resource Optimizer [Jira Board: ALKLE026]

1. Ready-to-Input KPIs (Project Tracking → KPIs Tab → + Add KPI)

KPI Name	Target	Unit	Sample Measurement	Evidence Note
Model Inference & System Latency	50	ms	42.8 ms	Triton Dynamic Batching benchmark logs on NVIDIA RTX 4090 cluster.
Lab Seat & GPU Utilization Efficiency	85	%	88.4 %	Campus telemetry MQTT broker aggregated report across 6 computing labs.
Pipeline Allocation Throughput	120	req/sec	135 req/sec	Locust stress test results showing zero dropped requests under peak load.

2. Ready-to-Input Challenges (Project Tracking → Challenges Tab → + Add Challenge)

Field	Challenge #1 (Demo Copy-Paste)	Challenge #2 (Demo Copy-Paste)
Title	High GPU memory contention during concurrent model inferencing	MQTT sensor heartbeat packets dropped under campus firewall rules
Root Cause	Multiple client microservices allocating unshared VRAM buffers without request throttling.	Outbound port 1883 blocked on primary academic VLAN subnet.
Impact	Inference latency spikes up to 340ms, causing timeouts in campus desktop app.	Real-time occupancy status displays 0 seats available in Lab 204.
Support Required	Server access to configure Triton Dynamic Batching with priority queues.	Campus Network Admin to whitelist port 1883 or enable WSS bridge.
Action Description	Deploy Triton dynamic queue manager and set maximum batch size to 8.	Switch sensor telemetry client to Secure WebSockets (port 443/WSS).
Action Owner	KLE Student Lead	Vihaan Hegde
Jira Sync Effect	Issue automatically created in Jira board ALKLE026 with Bug/Task type.	Issue automatically synced to Jira board ALKLE026 with student-blocker label.

Project: AL-KLE-023 — Adaptive Video Streaming & Buffer Management Engine [Jira Board: ALKLE023]

1. Ready-to-Input KPIs (Project Tracking → KPIs Tab → + Add KPI)

KPI Name	Target	Unit	Sample Measurement	Evidence Note
End-to-End Streaming Latency	50	ms	38.5 ms	WebRTC peer-to-peer data channel telemetry timestamp difference.
Buffer Stall / Rebuffering Ratio	0.5	%	0.32 %	ExoPlayer analytics telemetry under synthetic 3G/4G bandwidth throttling.
Frame Rendering Throughput	60	fps	59.4 fps	Hardware-accelerated OpenGL surface view frame counter log.

2. Ready-to-Input Challenges (Project Tracking → Challenges Tab → + Add Challenge)

Field	Challenge #1 (Demo Copy-Paste)	Challenge #2 (Demo Copy-Paste)
Title	Dataset preprocessing RAM overflow during video chunk indexing	Adaptive bitrate (ABR) oscillations during rapid network handoff
Root Cause	Loading uncompressed 4K raw video chunks directly into worker RAM.	Throughput estimation heuristic reacting aggressively to packet burst drops.
Impact	Training job crashes with Exit Code 137 (OOM Killer) at epoch 4.	Frequent quality downgrades causing visual buffering artifacts for viewers.
Support Required	Allocation of compute VM with PyTorch memory mapping support.	Faculty Mentor guidance on BOLA (Buffer Occupancy based Lyapunov) tuning.
Action Description	Implement streaming PyTorch chunk generators with disk memory-mapping (mmap).	Implement exponential moving average filter on throughput estimation window.
Action Owner	Aarav Kulkarni	Rohan Patil
Jira Sync Effect	Issue automatically created in Jira board ALKLE023.	Issue created and linked with Corrective Action in Jira board ALKLE023.